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Sex Differences in the Course of Personality Development: A Meta-Analysis

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Size and stability of sex differences in personality growth throughout adolescence and adulthood are examined. Studies using the Washington University Sentence Completion Test of ego development served as the primary source data. Sixty-five studies (encompassing more than 9,000 Ss) generated 113 independent effect sizes. Sex differences in ego development were moderately large among junior and senior high-school students (female advantage), declined significantly among college-age adults, and disappeared entirely among older men and women. Sex differences were relatively stable during early and middle adolescence. The greater maturity displayed by adolescent girls is not an artifact of superior verbal abilities: Sex differences in ego development were more than twice the magnitude of differences in vocabulary skills (Hyde & Linn, 1988). The present findings, together with findings from reviews of sex differences in moral judgment, aggression, and empathy, suggest that adolescent girls achieve developmental milestones earlier than boys, a difference that declines with age.

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Sex differences in personality development are not well documented beyond the common observation that young girls are less aggressive and less egocentric than young boys. Girls are typically portrayed as more mature than boys (Pitcher & Schultz, 1983), but it is not known if this sex difference extends past early childhood. It is not known if females consistently reach developmental milestones earlier than males, or if this advantage disappears with age or even reverses itself, revealing males as the more precocious sex. If sex differences in personality development do disappear, is the decline gradual or abrupt, and does it occur during early adolescence, late adolescence, young adulthood, or old age? Despite the burgeoning literature on sex differences, these questions have never been addressed. Indeed, despite 80 years of research (e.g., Eagly, 1987; Maccoby & Jacklin, 1974; Tyler, 1947; Wellman, 1933; Woolley, 1910), sex differences in temperament and social behavior have rarely been reviewed from a developmental perspective.

This gap in understanding has been due to the absence of a well-defined model of personality development that could be used to evaluate male and female development. The recent proliferation of stage theories has clarified the major milestones of growth, opening the way for male–female comparisons that have hitherto been absent. The most comprehensive developmental scheme is found in Loevinger's (1976) system of ego development, which supplies both a detailed model of development and a rigorous method for its assessment (the Washington University Sentence Completion Test [WUSCT]; Loevinger & Wessler, 1970; Loevinger, Wessler, & Redmore, 1970). The WUSCT has been used in more than 280 studies, translated into six languages, and administered to subjects of diverse ages and social backgrounds. These samples provide a large empirical base for estimating the size and stability of sex differences in personality development.

The purpose of this article is twofold. First, I review contemporary beliefs about sex differences in personality development and the extent to which these beliefs are supported by available evidence. This evidence has typically been indirect and inferred from studies of aggression, empathy, and moral judgment. Sex differences in these domains have been documented in recent meta-analyses, and those findings are reviewed here to generate tentative hypotheses about the size and stability of sex differences in personality development. These hypotheses need to be tested with evidence obtained from more direct and comprehensive assessments of personality development.

The second purpose of this article is to move beyond circumstantial evidence and present a meta-analysis of sex differences in personality development, in which WUSCT findings were used as the primary source data. Restricting the meta-analysis to WUSCT findings eliminated the most frequent pitfall of quantitative reviews: the inappropriate combining of findings derived from unrelated assessment procedures and conceptual schemes.

Definitions of Development

Investigators have described personality development in numerous ways. Gesell, Ilg, and Ames (1956) portrayed development as an increasing capacity for self-control, self-examination, and self-awareness. Block (1971) observed similar personality changes in his longitudinal study of development. Subjects in Block's study became more reflective with age, displayed an increasing concern with philosophical issues, and exhibited greater acceptance of responsibility for themselves and others.

Ausubel (1954) described a series of developmental achieve-

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ments that were similar to the advances outlined by Gesell et al. (1956) and Block (1971). Between childhood and late adolescence, observed Ausubel, individuals replace immediate desires with long-range goals, develop an ability to tolerate frustration, and adopt moral values because of their apparent validity rather than because of their approval by parents or peers.

Many of the age-related personality changes informally described by these early investigators resemble the advances depicted in current stage models of growth (Damon, 1977; Kohlberg, 1984; Selman, 1980), most notably Loevinger's (1976) model of ego development.

Loevinger's System

Loevinger (1976) has portrayed development as encompassing increasingly complex perceptions of the self and others. Conscious preoccupations move from the concrete to the abstract, and time orientation shifts from the immediate to the long term. The perception of other people, initially organized around simple dichotomies (e.g., nice vs. mean) becomes increasingly complex, encouraging a greater toleration of individual differences. These changes are accompanied by increasing psychological awareness and recognition of one's own internal motivations.

In Loevinger's (1976) system, personality development proceeds through a succession of stages. Each stage is characterized by a distinct developmental milestone or achievement.

Impulsive Stage

The earliest stage portrayed in Loevinger's (1976) developmental scheme is characterized by an absence of cognitive complexity and psychological insight. Perceptions of oneself and other people are organized around simple dichotomies, such as good versus bad and nice versus mean. Parents and peers are perceived as a constant source of supplies, reflecting the egocentric stance of this period. Individuals at this stage are oriented toward the immediate satisfaction of their wants, which leads to tantrums and other impulsive behaviors when their whims or desires are frustrated.

Self-Protective Stage

This stage is characterized by the emerging restraint of impulsive behavior. The emergence of self-control is motivated by a desire to avoid trouble and punishment, two preoccupying concerns of the self-protective period. This period is also characterized by a preoccupation with issues of control. Among children, this concern is expressed in the frequent wish to boss rather than be bossed; among adults, this wish is more manipulative and can lead to exploitive and opportunistic behavior.

The perception of oneself and other people continues to lack complexity, and emotions are understood with equal simplicity. One feels either happy or sad, and experiences are perceived as either fun or not fun. Such adjectives as *frustrating, interesting,* and *rewarding* are not spontaneously applied to oneself or others.

Conformist Stage

This stage is characterized by a declining concern with punishment and trouble and an increasing preoccupation with social acceptance and approval. Social interactions become less egocentric as individuals adopt conventional values and beliefs. For example, individuals at the conformist stage are the most likely to state that a wife should "accept her responsibility" or that a good father "tries to do the right thing" (Loevinger, Wessler, & Redmore, 1970; Redmore, Loevinger, & Tamashiro, 1978). The right thing is defined by social norms, which are perceived as universal and absolute.

Emotional life is described in banal and global terms. Such adjectives as *wonderful, important,* or *difficult* are used to describe activities, such as raising a family. These descriptions are less concrete than the responses found at previous stages (e.g., "is a bad problem," "a hard job," "takes money"), but they do not yet incorporate the contingencies and qualifications of thought revealed at later developmental levels (e.g., "Raising a family—is stressful but rewarding").

Self-Aware Level

This level represents a transition between the Conformist and Conscientious stages. It is characterized by an emerging self-awareness and capacity for introspection. Such adjectives as *self-conscious, lonely,* or *embarrassed* are used with greater frequency than at earlier stages. Social norms and rules are no longer viewed as immutable.

Conscientious Stage

This stage is characterized by the emergence of self-evaluated standards. Rules are followed because they are viewed as just, rather than because their violation leads to punishment or social disapproval. One's personality is regarded as more complex and differentiated than at previous stages. Such adjectives as *aggressive, materialistic, disorganized,* and *optimistic* are applied to oneself and to others, reflecting a greater awareness of emotional life. There is recognition that other people have their own interests, problems, and opinions, making it possible for one to tolerate and appreciate alternative viewpoints.

Individual and Autonomous Stages

These two stages are characterized by an increasing time perspective and increasing respect for individual differences. There is a growing appreciation of psychological causation (e.g., "If my mother—hadn't married so young [19] and had less than five children I may have looked forward to a family"; Loevinger, Wessler, & Redmore, 1970, p. 49). The Individual and Autonomous stages are also characterized by a preoccupation with issues of identity and emotional dependence and independence.

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Sex Differences

Stereotypic Beliefs

Girls are frequently portrayed as empathic, cooperative, and socially attuned, whereas boys are depicted as impulsive, aggressive, and egocentric (Langlie, 1937, and Terman, 1925, cited in Terman, 1946). Even among children, girls have a reputation for greater maturity and are perceived by their peers as more cooperative and altruistic than boys (Hartshorne, May, & Shuttelworth, 1930; Zarbatany, Hartmann, Gelfand, & Vinciguerra, 1985).

Early studies supported the validity of these stereotypes. Wickman (1928, cited in Wellman, 1933) found that young school boys were almost twice as likely as girls to be reported for bullying, rudeness, disobedience, and aggressiveness. Terman (1946) reviewed similar findings from studies of nearly 5,800 students, concluding that the findings were "so typical that it would be a waste of time to cite others" (p. 978).

The sex differences in maturity that are so clearly portrayed in descriptions of children are difficult to discern in descriptions of adults. Studies of gender stereotypes indicate that women are commonly portrayed as affectionate, sensitive, sympathetic, warm, gentle, cheerful, shy, and submissive, whereas men are pictured as aggressive, competitive, ambitious, dominant, rational, unemotional, independent, and self-confident (Ashmore, Del Boca, & Wohlers, 1986; Lueptow, 1984). Yet stereotypic feminine traits, such as gentleness and submissiveness, imply no greater or lesser capacity for impulse control or psychological insight than do masculine traits, such as independence and competitiveness. The characterization of such traits along bipolar personality dimensions (e.g., expressiveness–instrumentality) provides no additional insight into male–female differences in maturity: Expressiveness is no greater hallmark of development than is instrumentality.

By adulthood, males are no longer portrayed as the less mature sex. Indeed, perceptions of maturity may actually undergo a reversal. Sheriffs and McKee (1957) reported that college students are more likely to associate the term *maturity* with men than with women. Until recently, the accuracy of these gender stereotypes had not been evaluated.

Empathy

The capacity to adopt another person's perspective is one of the hallmarks of personality development. Because females are typically portrayed as more empathic and sensitive than males, females are assumed to have a greater capacity for understanding the thoughts and feelings of other people.

Hoffman (1977) reviewed the evidence for sex differences in two types of empathic responses. The first type of empathic response was defined as a "cognitive awareness of another person's feelings" (p. 712). Individuals with this ability can provide accurate descriptions of someone else's feelings and thus display a capacity for perspective taking. The second type of empathic response was defined as a vicarious affective reaction to another person's situation. People with this ability can mimic the emotional response of a target person. Babies, for example, might cry upon hearing the cries of another infant, or adults might report feelings of distress while watching a confederate receive shocks.

Hoffman (1977) examined the evidence for sex differences in empathic abilities in 14 samples of children (N = 1,297; ages 3–11). Hoffman found no consistent sex differences in children's ability to identify the affective state of another person, suggesting that boys and girls do not differ in their capacity for perspective taking.

Hoffman (1977) did find sex differences in the ability of newborns and young children to mimic the emotional response of a target person. In each of the 15 samples reviewed, girls obtained higher affective empathy scores than boys, although significant or marginally significant differences were obtained in only 6 samples. Hoffman concluded that boys and girls are equally capable of assessing how someone feels but that in girls the "awareness of the other's feelings is more apt to be accompanied by a vicarious affective response" (p. 716).

Aggression

The capacity to control the expression of immediate impulses represents a second aspect of personality development. Males are typically portrayed as more aggressive than females, perhaps indicating poorer impulse control in boys and men.

Gender differences in aggression have been documented in two recent meta-analyses. Hyde (1984, 1986) reviewed findings from 143 studies of nonclinical populations ranging in mean age from 2 to 35. As expected, males in these studies were significantly more aggressive than females. The size of the gender difference depended on a number of study characteristics, including the age of the sample. Gender differences were large among 4- to 5-year-olds, moderate among 9- to 12-year-olds, and small among college-age adults (mean d = 0.86, 0.54, and 0.27, respectively). Only one sample of teenagers (age 12) was included in the analysis, so no conclusions can be drawn about the size or stability of gender differences in aggression during early and late adolescence.

The decline in gender differences between early childhood and early adulthood may be due to males' increased capacity to control aggressive impulses, or the outcome may be an artifact of the different procedures used to assess aggression in children and adults. Among preschoolers, investigators measured sex differences in physical aggression, whereas among college students investigators measured sex differences in the willingness of subjects to shock another person.

Eagly (1987; Eagly & Steffen, 1986) conducted a quantitative review of 63 studies of aggression in adolescents (age 14 or older) and adults. Consistent with Hyde's (1984, 1986) review, males were found to be more aggressive than females. Here, too, the size of the gender difference depended on a number of study characteristics, including the type of aggression assessed (psychological vs. physical) and the privacy of the aggressive act (i.e., no privacy, semiprivate, private). Age effects were not examined. The largest gender difference was found in studies of physical aggression (Hedges' weighted mean d = 0.40); the smallest gender difference was found in studies in which subjects believed that their behavior was unobserved (Hedges weighted mean d = 0.17). Both reviews support the conclusion

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that males are more aggressive than females, a difference that is small by early adulthood.

Moral Development

The growth of moral principles is a third component of personality development. Some early investigators perceived an initial female advantage (Gesell et al., 1956; Piaget, 1932/1965), whereas other investigators perceived a male advantage (Freud, 1925/1976).

Recent debates have focused on a possible gender bias in Kohlberg's (1984) theory and measure of moral development. Kohlberg depicted six stages in the development of moral reasoning. Moral judgments at Stage 1 and Stage 2 are characterized by concrete and egocentric concerns. Judgments at Stages 3 and 4 are characterized by a social perspective, with moral decisions based on interpersonal considerations (Stage 3) and abstract societal considerations (Stage 4). Judgments at Stages 5 and 6 recognize the relative nature of many moral positions. Stage assignment is based on a person's responses to a series of hypothetical moral dilemmas. These responses are coded for developmental level by using a published scoring manual (Colby et al., 1987).

Most adults progress beyond the first two stages of moral judgment. Early studies suggested that women progress to Stage 3 and men progress to Stage 4, leading Gilligan (1982) to question the validity of Kohlberg's system for evaluating the development of moral reasoning in women.

Walker (1984) reviewed the evidence for sex differences in moral reasoning in 79 studies in which Kohlberg's Moral Judgment Interview (MJI) was used. No sex differences were found when Stouffer's method for combining Z scores ($Z = 0.73$, $p = .23$, one-tailed). Walker found sex differences in only 6 of 41 samples of children and adolescents ($N = 2{,}879$); girls obtained higher MJI scores in 5 of those samples. Among 46 samples of high-school and university students ($N = 3{,}901$), there were again only 6 significant sex differences: 5 of these favored men. Finally, among 21 samples of adults (ages 21–65, $N = 1{,}223$), there were 4 significant differences, all favoring men. Walker's analyses do not support allegations of sex differences in the growth of moral reasoning. However, he did not provide effect-size estimates or related statistics that might lend greater strength to this conclusion.

Thoma (1986) reviewed findings of sex differences among studies in which Rest's (1979) Defining Issues Test (DIT) was used. The DIT is a paper-and-pencil measure of moral judgment derived from Kohlberg's (1984) theory of development. The DIT presents subjects with a series of moral dilemmas, followed by a list of issues that might be considered important for resolving the dilemmas. Respondents choose four issues that they judge to be most important in the resolution of each dilemma. Each choice is assigned to a developmental level according to an a priori coding scheme. Respondents score about half a stage higher on the DIT than on Kohlberg's MJI (Rest, 1979).

Thoma (1986) analyzed findings from 54 studies in which the DIT was used. Gender differences favored females among junior and senior high-school students, college youth, and adults (mean $d = .15, .17, .21$, and $.28$, respectively). Effect sizes did not vary significantly among age groups, suggesting that sex differences on the DIT are small and stable.

Both reviews (Thoma, 1986; Walker, 1984) reveal a tendency for adolescent girls to obtain slightly higher moral reasoning scores than adolescent boys. This outcome is consistent with conventional wisdom but unexpected from the standpoint of contemporary theory, which predicts either no sex differences or a small difference favoring males (Gilligan, 1982).

The fate of sex differences among adults is not clearly depicted in these two reviews. Thoma (1986) reported a continued female advantage, whereas Walker (1984) found an absence of sex differences. Because responses to the DIT may rely more heavily on verbal facility than do responses to Kohlberg's MJI interview (Thoma, 1986), women's superior performance may be an artifact of their slightly greater verbal abilities (Hyde & Linn, 1988).

Summary

When these findings are viewed together, females appear more mature than males, but the advantage is small at each age level (see Table 1). When defined as the ability to accurately label the feelings of another person, perspective taking appears no greater in girls than in boys. The complexity of moral judgment also seems similar among both sexes, although girls seem slightly less concrete and egocentric in their reasoning than did boys. Adult women display moral judgments on the DIT that are slightly more advanced than those of their opposite-sex peers. Aggression is the only domain in which sex differences are initially large, and here too male–female differences shrink by early adulthood.

In summary, these findings suggest that gender differences in personality growth are small, favor females, and are relatively stable between childhood and young adulthood. However, the evidence generating these hypotheses is limited in three respects. First, the evidence is fragmentary and drawn from only a few dimensions of personality development. Convincing evidence for sex differences must come from studies in which more comprehensive assessment procedures are used. Second,

Table 1
Magnitude of Sex Differences in Aggression, Empathy, and Moral Judgment

	Effect size (mean d)		
Age of subjects	Aggression	Empathy	Moral judgment
4–5	−0.86	*ns*	—
9–12	−0.54	*ns*	—
12–17[a]	—	—	0.15–0.17
18–22[b]	−0.27	—	.21
>22	—	—	.28

Note. Effect sizes were obtained from Hoffman (1977), Hyde (1984), and Thoma (1986). Positive values indicate a female advantage. Dashes indicate that no data are available.
[a] Junior and senior high-school samples. [b] College samples.

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the evidence provides little insight into the size and stability of sex differences during early, middle, and late adolescence. Few samples of teenagers were included in the quantitative reviews, so no conclusions can be drawn about gender differences during this formative period. Finally, the most dramatic sex difference, greater aggression among males, may not be a valid sign of sex differences in impulse control or, more generally, development. Male aggressiveness may reflect the demands of social roles (Eagly, 1987) or the internalization of cultural expectations and values during childhood (Maccoby & Jacklin, 1974). If either interpretation is correct, then sex differences in maturity may be smaller than currently believed. Using WUSCT findings as the primary source data, I examined more precisely the size, stability, and direction of sex differences in personality development.

Method

Measurement of Ego Development

The WUSCT is a semiprojective measure composed of 36 sentence stems (Loevinger & Wessler, 1970). Respondents are requested to complete each stem; no additional instructions are usually given. The ensemble of responses reveals a person's characteristic mode of reasoning about his or her actions, motivations, and personal relationships. Protocols that display little time perspective, impulse control, or self-reflectiveness are scored at the low end of the developmental continuum. Protocols revealing a capacity for psychological insight and cognitive complexity are scored at higher levels of the continuum.

An empirically derived scoring manual is used to assign each response to a developmental level (Loevinger & Wessler, 1970; Loevinger, Wessler, & Redmore, 1970; Redmore et al., 1978; Loevinger, Lê-Xuân-Hy, & Associates, 1989a, 1989b, 1989c). The final distribution of 36 item ratings is converted into a single composite score, or total protocol rating (TPR), by using an algorithm provided in the scoring manual. After completing individual item ratings, investigators may also read the protocol in toto to identify responses that merit either higher or lower ratings when understood in the context of the whole protocol. The final TPR may be adjusted for this additional information.

An item-sum scoring algorithm provides an alternative method for deriving a single score. This method has the apparent advantage of increased discrimination and thus may be more sensitive to small differences between groups of individuals under study. However, item-sum ratings are insensitive to signs of ego development that emerge only from reading the protocol in toto, for example, the variability of themes displayed at high levels or the redundancy of themes often found at low levels. Moreover, respondents at the highest stages of development may be more aware of and willing to reveal concerns that cut across a number of developmental levels. Respondents at slightly lower ego stages are less aware of and less willing to reveal these concerns, yet the item-sum score may award higher ratings to their protocols (Holt, 1980; Loevinger & Wessler, 1970).

Internal consistency of the WUSCT is high. Loevinger reported a value of .91 for Cronbach's alpha (Loevinger & Wessler, 1970). Holt (1980) administered a 12-item form of the WUSCT to a national probability sample of 966 adults and obtained almost identical results when corrected for length. Other investigators have reported similar values (e.g., Hansell, Sparacino, Ronchi, & Strodtbeck, 1985). Interrater reliability is also reported to be high (Dubow, Huesmann, & Eron, 1987; Gfellner, 1986; Loevinger & Wessler, 1970).

WUSCT forms exist for males and females. Although the scoring manual was originally developed with female subjects, a supplementary scoring manual for boys and men has also been constructed (Redmore et al., 1978).

The test has proceeded through a series of changes, with some items discarded over time and others added. In its complete form, the WUSCT consists of 36 sentence stems. Form 81 for women and Form 81 for men include the most recent revisions of the test and are closely comparable for both sexes (Loevinger, 1985). For reasons of experimental design, the test is sometimes administered in an abbreviated format. Form 81 has been constructed so that both halves of the test can be used as 18-item measures when necessary.

Form 81 for women and Form 81 for men are virtually identical (Loevinger, 1985). These forms contain 30 identically worded sentence stems; 3 additional stems differ because of a change in pronoun (e.g., "At times she (he) worried about—"). The remaining three sentence stems are similar in content and differ only in terms of the gender of the subject or direct object of the sentence (e.g., "A woman (man) should always—"). The correlations of individual item ratings with the TPR (i.e., item validities) are comparable for the male and female forms (*Mdn* = .46 and .50, respectively; Loevinger, 1985).

Form 11–68 for women and Form 11–68 for men were used in early studies of ego development. These forms are closely comparable for both sexes and contain 18 identically worded items. An additional 4 items differ only in the gender of the pronouns (e.g., "Sometimes she (he) wished that—"). Six additional stems differ slightly in content because of a gender change in the subject of the sentence (e.g., "A woman (man) should always—"). Eight items contain no overlapping content. Item validities are slightly higher on the female form than on the male form (median validity for first-person stems = .50 and .425, respectively; Loevinger, 1985).

Separate forms exist for use with children. Boys and girls as young as 12 years of age appear to have sufficient writing skills to complete the test without difficulty (McCammon, 1981). Here, too, male and female forms are closely comparable. Form 2–77 for boys and Form 2–77 for girls contain 25 identically worded sentence stems, and an additional 4 stems differ only in the pronouns used (e.g., "At times he (she) worried about—"). Five additional sentence stems differ slightly in content because the gender of the subject or direct object of the sentence varies (e.g., "A good father (mother)—").

Construct Validity

The construct validity of the WUSCT has been evaluated in previous reviews (Hauser, 1976; Loevinger, 1979). As predicted, low ego levels are associated with impulsiveness (Starrett, 1983), adolescent pregnancy (Hart & Hilton, 1988), authoritarianism (Browning, 1983), and delinquent behavior (Frank & Quinlan, 1976). Assessments of conformity show the predicted curvilinear relationship with WUSCT scores (Hoppe & Loevinger, 1977), with individuals at the Conformist level displaying the greatest degree of behavioral conformity. Empathy, defined as the ability to detect and describe the affective state of a target person, is also associated with ego level in the predicted direction (Carlozzi, Gaa, & Liberman, 1983). Summarizing the evidence for external validity, Loevinger (1979) observed that "at low ego levels there is behavioral evidence confirming the syndromes; at higher levels differential evidence lies in attitudes and ideas" (p. 281).

The relationship between ego level and moral reasoning has also been in the predicted direction. Lee and Snarey (1988) reviewed findings from nine studies in which both the WUSCT and Kohlberg's MJI interview were used. Complete data sets were obtained for each study and combined into a single composite sample (N = 787). The correla-

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tion between measures was high ($r = .65$, $p < .001$) but probably inflated because of the age range among subjects (ages 11 to 82).

Literature Search

A preliminary data base was generated by conducting a computer search of *Psychological Abstracts* from 1970 (the publication date of the scoring manual) through 1983. A manual search was conducted for the years 1984 through 1987 (April). The key words employed were *Loevinger, Washington University Sentence Completion Test,* and *ego development.* Additional studies were identified by using previously compiled bibliographies (e.g., Holt, 1980) and citations in review articles (Hauser, 1976; Loevinger, 1979). Unpublished studies were located with a manual search of *Dissertation Abstracts International* (1977–1986) and by informal inquiries.

These search procedures located 112 published studies and 165 unpublished studies in which the WUSCT was used. From this population, 45 published studies and 98 unpublished studies were excluded, leaving a potential pool of 134 studies for inclusion in the meta-analysis. Studies were excluded from the analysis for the following reasons:

1. Subjects were all of the same gender ($n = 88$).
2. Subjects were drawn from clinical populations (e.g., hospitalized psychiatric patients or drug addicts; $n = 10$).
3. Findings were reported in previously published articles ($n = 11$).
4. Sampling procedures were inappropriate for male and female comparisons (e.g., subjects were chosen to represent a range of ego levels within each sex; $n = 8$).
5. Protocols were not rated with the scoring manuals ($n = 1$).
6. Age differences between boys and girls were present (Barglow et al., 1983) or might have been present (Block, 1973; Lorr & Manning, 1978; $n = 3$). This criterion was not applied to adults older than 22 because ego development appears to stabilize by the early 20s, making age differences between males and females an unlikely confound when examining gender differences in ego development.
7. Sampling demographics were not provided ($n = 23$).

Deficient Reporting

Requests for additional data were sent to investigators when information was missing from published reports. Authors of dissertation studies were also contacted when their addresses could be obtained from either the American Psychological Association or National Faculty directories. Seventy-one investigators were initially contacted by mail (27 authors of published studies and 44 authors of unpublished studies), and an additional 40 follow-up letters were sent to nonresponders. Letters usually contained requests for the following information: distributions of male and female ego levels (by age group, when appropriate), mean item-sum scores and associated standard deviations for males and females, mean ages for males and females, the scoring key used for converting ego level ratings into numerical values, number of raters per protocol, and "any additional information that might bear on a comparison of male and female ego levels." Thirty-four (48%) investigators replied to these inquiries: 24 provided information that was useful for the analysis.

In many studies, subjects' grade level was reported but not their age. In these cases, grade level was converted into age with the following scale: sixth grade = 11–12, seventh grade = 12.1–13, eighth grade = 13.1–14, ninth grade = 14.1–15, tenth grade = 15.1–16, eleventh grade = 16.1–17, and twelfth grade = 17.1–18.

Meta-Analysis Techniques

The index of effect size used in the meta-analysis was Hedges' (1982) unbiased g (g^U), where g represents the mean difference between male and female ego scores divided by the pooled within-groups standard deviation. Hedges has shown that g is a biased estimator of the population effect size. Multiplying g by a constant corrects for this bias. This procedure is similar to multiplying a sample variance by a correction term to obtain an unbiased estimate of the population variance.

Whenever possible, effect sizes were derived from TPR scores ($n = 94$). Eleven effect sizes were derived from item-sum scores.

Eleven studies did not provide sufficient information to calculate g^U directly but did provide the exact results of a t test for gender differences. In these cases, effect sizes were computed by transforming t into Cohen's d (Cohen, 1977). Point-biserial correlations between gender and ego level were also transformed into estimates of d.

Four studies did not provide either descriptive statistics or test statistics but did indicate if gender differences were present. In these cases, the convention was adopted of setting t to the smallest value that would yield a significant finding given the particular sample size and alpha equal to .05. When no significant gender difference in ego level was reported, then the t value was set at zero and the associated p value at .50 (Strube & Hartman, 1983).

A second index of effect size, weighted g^U, was also computed for each study. This index weights each g^U value by the inverse of its variance (Hedges, 1982). Because the variance of an effect size is largely a function of its sample size, this procedure adjusts for sample-size differences between studies.

Combining effect sizes across studies is only appropriate when effect-size estimates are homogeneous. A test for heterogeneity among estimates is provided by Hedges (1983; Hedges & Becker, 1986). His test statistic, H_T, represents the sum of squared deviations around the grand mean of all effect sizes. Under the assumption of homogeneity, the test statistic H_T is distributed approximately as a chi-square distribution with $k-1$ degrees of freedom, where k represents the number of samples. A significant value of H_T suggests that variability among effect sizes is not due to sampling variance but rather the presence of at least one moderator variable.

Stouffer's method for adding standard normal deviates was used to assess the significance of all studies combined (Rosenthal, 1978). For each t value, the associated one-tailed p value was computed and transformed into a Z score. The sum of the Z scores was divided by the square root of the number of samples tested.

Results

The results of the meta-analysis are presented in Table 2. Sixty-five studies yielded 113 independent effect-size findings. Positive values indicate a gender difference in favor of females. A significant female advantage was found in 39 samples, and another 11 samples produced marginally significant ($p < .10$) results. A significant difference in favor of males was found in only 2 samples. Obtaining this number of significant findings when the population effect size is zero is unlikely to have occurred by chance (Stouffer's $Z = 12.16$, $p < .0000001$).

Effect sizes (g^U) ranged from −0.69 to 1.66. There was significant nonhomogeneity among effect sizes ($H_T = 246.5$, $p < .01$), which is probably due to the presence of at least one moderator variable.

Age was the most likely variable influencing the magnitude of gender differences. Because some samples were composed of subjects of equal ages, and other samples composed of subjects of unequal ages, two separate meta-analyses were conducted: one for homogeneous age groups and the other for heterogeneous age groups. This permitted a precise examination of the course that gender differences follow over time.

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Table 2
Study Characteristics

Study	Mean age	Grade	Sample size[a] Females	Sample size[a] Males	t	One-tailed p	g^U
Hirsh (1979)	11.7	5–6	56	37	−1.99	−.97	−0.41
Trimble (1984)		5–7	57	57[b]	1.98	.05	0.37
Blasi (1971)	12.6	6	44	32	0.21	.42	0.05
	12.6	6	22	25	−0.69	.753	−0.20
Sullivan, McCullough, & Stager (1970)[c]	12		22	22	0.66	.26	0.20
	14		26	23	−0.27	.606	−0.08
	17		19	26	−1.08	.847	−0.33
Cohn (1984)[d]	13.4	7	21	25	1.81	.04	0.54
Gfellner (1986)	12.7	7	63	54	3.17	.00099	0.59
	13.7	8	54	65	5.13	.0000007	0.94
	14.6	9	49	50	3.06	.0016	0.61
	16.5	11	37	25	2.05	.023	0.53
	17.5	12	14	10	1.55	.0712	0.65
Coor (1970)	13.2	7	14	14	1.12	.146	0.43
	14.3	8	8	10	2.83	.0018	0.14
	14.7	9	10	9	.75	.236	0.35
	16.0	10	9	9	1.57	.056	0.75
Nichols (1981)[e]		7	31	26	−2.61	.994	−0.69
Starrett (1983)	13.6		27	26	4.22	.00005	1.16
	16.9		28	17	3.42	.00073	1.06
	18.2		46	19	1.80	.04	0.49
Head & Shayer (1980)	14		79	84	3.38	.00045	0.53
	17		32	33	0.00	.50	−0.01
J. Block (1987)	14		53	51	−0.33	.629	−0.06
Loevinger & Cohn (1980)		9	17	15	1.85	.039	0.66
Hansell (1981)[f]		9	29	28	6.25	.0000001	1.66
		10	33	24	2.66	.005	0.72
		11	29	26	2.59	.0065	0.70
		12	25	21	0.68	.253	0.20
Redmore & Waldman (1975)	14–16	9	25	26	2.38	.011	0.67
	18–37		41	40	−0.40	.654	−0.09
Hauser, Jacobson, Noam, & Powers (1979)	14		22	26	2.10	.022	0.61
	14.5		42	34	1.91	.032	0.44
	13.5		142	152	4.18	.00004	0.49
Tamashiro (1979)	16.1		64	55	3.16	.001	0.58
	34.3		24	11	0.77	.226	0.28
Brownlee (1985)	16		89	64	3.38	.00046	0.56
Kitchener, King, Davison, Parker, & Wood (1984)		11	10	9	1.18	.133	0.55
		15	18	16	−0.32	.625	−0.11
		17+	9	10	1.14	.141	0.53
Schenberg (1974)	17.1		74	37	0.00	.50	0.00
Brustin (1984)		12	60		2.79	.0038	0.72
Hurtig, Petersen, Richards, & Gitelson (1985)	17–18	12	72	67	4.59	.0000057	0.78
Lamb (1971)	14–19		49	56	3.59	.000257	0.70
Lambert (1972)	11 & 15		32	27	1.83	.038	0.48
Lerner (1984)	11–20		28	31	1.62	.057	0.42
Hood (1987)		13	61	36	0.33	.368	0.07
Adams & Fitch (1982)		13	28	10	0.45	.33	0.16
		14	28	23	0.20	.42	0.06
		15	23	27	−0.62	.728	−0.18
Candee (1974)		15	76		0.00	.50	0.00
White, Speisman, & Costos (1983)	24.1		78	74	1.40	.086	0.23
Alishio & Schilling (1984)	20		31	29	0.66	.255	0.17
Nelson (1980)	33.1		10	10	0.75	.236	0.34
	39.4		10	10	−0.76	.7668	−0.34
	49.6		10	10	0.27	.396	0.12
Dubow, Huesmann, & Eron (1987)	30–31		206	192	2.91	.00206	0.29
Weis & Dain (1979)	30.1		50	50	1.34	.093	0.27
	31.7		50	49	−0.87	.7972	−0.17
Hauser, Powers, Noam, Jacobson, Weiss, & Follansbee (1984)	43.5		61	61	−0.96	.8196	−0.17
Swensen, Eskew, & Kohlhepp (1981)	61		36	36	0.00	.50	0.00
Streich & Swensen (1985)	20.5		32	32	0.10	.46	0.03
	40.9		32	32	3.68	.0003	0.92
	73.3		32	32	0.00	.50	0.00
Looney (1985)	19.6		30	30	0.90	.195	0.23
Redmore (1976)	18–23		22	20	1.99	.027	0.62
Browning (1987)[h]	16–18		178	180	0.50	.308	0.05
	19–21		158	153	2.91	.002	0.33
	22–25		139	122	1.64	.052	0.20
Rosen & Nordquist (1980)	>21		15	13	−0.20	.58	−0.07
Bailey & Cohn (1982)	14.8		32	24	3.24	.002	0.88
McCammon (1981)		6 & 10	40	40	3.83	.00021	0.86
Snarey, Friedman, & Blasi (1986)	14		10	10	2.46	.013	1.10
	21–42		10	10	1.10	.154	0.49
Fischer (1973)		6–8	47	46	0.79	.223	0.16
West (1973)	23.6		42	56	0.82	.215	0.17
Goldsmith (1971)		17+	10	47	0.75	.233	0.26
Nettles & Loevinger (1983)	36.3		107	107	0.00	.50	0.00
Streich (1987)[i]	41.6		21	21	2.03	.025	0.63
Valentino (1978)	44.2		115	115	−0.55	.705	−0.04
Martin (1983)	41.5		25	25	−0.84	.7879	−0.24
Fleming, Costos, & MacGowan (1984)[j]	31.8		22	22	−0.55	.705	−0.16
DeLoach (1976)	15–44	15.2	21	20	0.00	.50	0.00
		15.2	15	6	−0.76	.7673	−0.36

Whiteley, Bertin, Jennings, Lee, Magana, & Resnikoff (1982)[g]	13	18	16	2.15	.021	0.74	Butler (1971)	34.5	68	74	1.31	.097	0.22
	13	20	19	−0.20	.58	−0.06	Suzman (1974)	25–45	202	70	0.87	.194	0.12
	13	25	21	1.13	.141	0.34	Hansell, Sparacino, Ronchi, & Strodtbeck (1985)	13–91	112	109	2.09	.02	0.28
	13	16	13	0.16	.44	0.06							
McAdams, Booth, & Selvick (1981)	13	6	5	0.32	.38	0.19	Schwarz & Robins (1987)	38.8	44	40	0.87	.195	0.19
	14	8	7	0.13	.45	0.07							
	15	8	8	1.88	.042	0.95	Levertov (1985)	Junior high through college	120		1.98	.025	0.36
	16	8	6	−0.78	.7684	−0.42							
Loevinger, Cohn, Bonneville, Redmore, Streich, & Sargent (1985)	13	26	30	−0.83	.7878	−0.22							
	13	14	30	0.57	.287	0.19							
	13	23	67	0.98	.176	0.24	McAdams, Ruetzel, & Foley (1987)	39.6	30	20	0.68	.252	0.20
	13	35	49	−0.63	.7323	−0.14							
	13	14	52	1.30	.099	0.39	Borders, Fong, & Neimeyer (1986)	29.9	45	18	0.50	.31	0.14
	13	62	76	1.37	.089	0.24							
	13	34	63	4.01	.000064	0.86	Acklin (1985)	15–60+	60	60	*ns*	.50	0.00
	13	17	46	1.54	.067	0.44							
	14	25	26	0.44	.3322	0.12	Pazy (1985)	19–50	57	50	−1.51	.93	−0.29
	16	42	37	2.55	.0067	0.58	Logan (1985)	17–27	77	53	1.37	.089	0.24
							Pate (1986)		95	95	1.93	.029	0.28

[a] When this information could not be obtained, sample sizes for males and females were assumed to be equal for computational purposes. [b] Sex differences were reported for Grades 1, 3, 5, and 7, but computations were based only on Grades 5 and 7. [c] These data were obtained from Jane Loevinger. [d] Only Grade 7 data were available for analysis. [e] Only Grade 7 (nonlongitudinal) data was used in the analysis. [f] Mean ego scores for males and females at each grade level were obtained from ERIC document #174719 (Hansell, 1979). Standard deviations for each grade were unavailable. Effect size computations are based on the standard deviation provided for all grades combined (Hansell, 1981). [g] Standard deviations were not provided for each sex, so computations are based on information provided for boys and girls combined. [h] The data reported in this study were weighted in a way that makes tests of sex differences problematic (R. Holt, personal communication, September 24, 1985). [i] Only staff data were used. [j] Heterosexual data only.

Homogeneous Age Groups

Eighty samples (N = 5,287) were distributed among 11 age groups. Six of these groups displayed significant variability among effect sizes ($ps < .05$). In these cases, extreme values (outliers) were identified and excluded from subsequent analyses. Although the number of outliers eliminated from each group was small (see Table 3, column 7), they suggest the presence of more than one moderator variable.

Gender differences were expected to be larger among adolescents than among adults. Results of the meta-analysis support this expectation (Table 3), with one exception: No significant gender differences were found among sixth graders. Because of the small number of samples in the analysis, one cannot determine if this finding is due to chance or a genuine absence of sex differences among sixth graders.

Junior and senior high-school girls scored significantly higher than boys on the WUSCT. Effect-size estimates (weighted g^U) ranged from 0.48 (7th graders) to 0.70 (12th graders). Inspection of the confidence intervals (CI) indicated that all values were significantly different from zero.

The fluctuation in effect-size values among samples of adolescents may reflect sampling variance or a systematic relationship with age. The relationship between age and effect size can be tested directly by using Hedge's (Hedges & Becker 1986) analog to analysis of variance (ANOVA). This procedure tests the significance of between-group variation among effect sizes. The analysis follows the same logic underlying ANOVA and entails partitioning the overall homogeneity statistic (H_T) into two component parts: between-group variation (H_B) and within-group (H_W) variation. The magnitude of H_B is tested against the desired critical value of a chi-square distribution with $p - 1$ degrees of freedom, where p equals the number of levels in the grouping factor.

In the present analysis, grade was used as the grouping factor (Grades 7, 8, 9, 10, 11, and 12), and effect size served as the dependent variable. Between-group variation was not significant ($H_B = 1.83$, $df = 5$, $p > .50$), suggesting that gender differences in ego development were relatively stable among these samples of adolescents. On average, adolescent girls scored three fifths of a standard deviation higher than boys on the WUSCT (weighted $g^U = 0.61$, CI = 0.52–0.70). Expressed in an alternative format, the mean ego level of adolescent girls was higher than the ego levels of 73% of adolescent boys (Cohen's $U_3 = 72.9$).

Among samples of college freshmen, gender differences declined dramatically. The mean effect size for freshmen was significantly smaller than the mean effect size for 9th graders ($Z = 2.50$, $df = 1$, $p < .025$), 10th graders ($Z = 2.34$, $df = 1$, $p < .025$), 11th graders ($Z = 2.34$, $df = 1$, $p < .025$), and 12th graders ($Z = 3.22$, $df = 1$, $p < .005$). This sudden decline cannot be attributed to the beneficial effects of college because at least 81% of the subjects completed the WUSCT in the fall of their freshman year. The decline may reflect sampling differences between high school and college subjects, a point returned to in the Discussion section.

Although male–female differences declined by the first year of college, women still displayed slightly greater maturity in

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Table 3
Mean Effect Sizes and Related Values, Broken Down by Age

Grade/age	Weighted g^U	H	95% CI for g^U	No. of samples	N	No. of effect sizes discarded[a]
			Homogeneous age groups			
5th–6th	−0.20	2.18	−0.46–0.07	3	216	1
7	0.48	5.18	0.22–0.74	4	245	1
8	0.63	7.73	0.47–0.78	5	677	3
9	0.60	0.76	0.36–0.84	5	277	1
10	0.63	0.24	0.34–0.92	3	194	0
11	0.58	0.25	0.34–0.82	4	289	0
12	0.70	4.25	0.47–0.93	5	304	3
Freshman	0.25	19.56	0.12–0.38	16	997	0
Sophomore–senior	0.13	9.29	−0.07–0.33	9	386	0
22 < mean age < 35	0.21	5.62	0.09–0.33	10	1,174	0
38 < mean age < 74	−0.07	1.10	−0.24–0.10	6	528	1
			Heterogenous age groups			
12–15	.55	8.14	0.34–0.76	6	367	0
16–18	0.19	8.17*		2	463	0
18–22	0.22	7.38	0.09–0.35	11	965	0
23–43	0.12	14.06	0.02–0.22	13	1,563	0

Note. H = homogeneity statistic; CI = confidence interval; N = number of subjects.
[a] The effect sizes discarded were from the following studies: Grade 6—Trimble (1984), $g^U = 0.37$; Grade 7—Nichols (1981), $g^U = -0.69$; Age 14—Block (1987), $g^U = -0.06$; Sullivan, McCullough, & Stager (1970), $g^U = -0.08$, and Coor (1970), $g^U = 0.14$; Grade 9—Hansell (1981), $g^U = 1.66$; Grade 12—Head & Shayer (1980), $g^U = -0.01$; Schenberg (1974), $g^U = 0.00$; and Hansell (1981), $g^U = 0.20$; 38 < mean age < 74—Streich & Swensen (1985; subjects' age = 40.9), $g^U = 0.92$.
* $p < .05$.

their sentence completions than did their male counterparts (weighted $g^U = 0.25$, CI = 0.12–0.38). After the first year in college, gender differences in ego development disappeared almost entirely. The effect size for sophomores, juniors, and seniors was small in magnitude and not significantly different from zero (weighted $g^U = 0.13$, CI = −0.07–0.33).

Among postcollege adults (22 < mean age < 34) there was a small but significant gender difference (weighted $g^U = 0.21$, CI = 0.094–0.33). This difference disappeared among men and women in the oldest age group (weighted $g^U = -0.07$, CI = −0.24–0.10). Hedges and Becker's (1986) analog to ANOVA was used to test the overall relationship between age and effect size among adults. There were four levels to the grouping factor (college freshmen, sophomores through seniors, 23 < mean age < 34; 39 < mean age < 73); effect size served as the dependent variable. Between-group variation was significant ($H_B = 9.22$, $df = 3$, $p < .03$), suggesting that gender differences among adults were not stable but declined with age. Post hoc comparisons revealed that the mean effect size of older adults (mean age > 37) was significantly smaller than the mean effect size of college freshmen, $Z(3) = 3.0$, $p < .05$, and 22- to 25-year-olds, $Z(3) = 2.7$, $p = .06$.

Heterogeneous Age Groups

The 32 samples (N = 3,358) defined as heterogeneous were distributed among four age groups: young adolescents (mean age < 16), older adolescents (ages 16–18), college students (freshman through senior years), and adults (mean age > 22).

The mean effect size for each group is presented in the bottom half of Table 3. Young adolescent girls scored significantly higher than boys on the WUSCT (weighted $g^U = 0.55$). This gender difference shrank dramatically among college students (weighted $g^U = 0.22$) and then disappeared almost entirely among older adults (weighted $g^U = 0.12$).

This pattern of findings is virtually identical to the pattern observed among homogeneous samples, where the mean effect size for all young adolescents (Grades 6–10), all college students, and all adults (mean age > 22) was 0.56, 0.21, and 0.12, respectively.

Stage Differences

To interpret the meaning of an effect size, it is necessary to return to the units of the original measure, in this case the mean ego scores of males and females. This information was recovered from 60 samples of males and females, representing findings from 4,598 individuals.

The majority of young adolescents fell between the Self-Protective and Conformist stages of development, with girls tending to score at least one stage higher than boys on the WUSCT (e.g., self-protective–conformist vs. self-protective, respectively). Girls appeared less egocentric and concrete in their thinking than boys and somewhat more concerned with social approval and acceptance. The majority of older adolescents fell between

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the Conformist and Self-Aware levels of development. Here, too, girls tended to score at ego levels at least one level higher than boys, implying greater movement toward emerging self-awareness and reflection. Male and female first-year college students tended to score between the Conformist and Conscientious stages of development. There was a small tendency for young women to be more reflective than young men.

Threats to Validity

Sampling bias: adults. Forty percent of the noncollege adult samples were composed of married couples or couples living together. This sampling bias may have minimized the presence of gender differences among adults because of homogamous mating. To test this possibility, I classified adult samples as either couples (husband–wife pairs or couples living together) or noncouples. There were 10 samples ($N = 982$) in the matched group and 15 samples ($N = 2{,}153$) in the nonmatched group.

Gender differences among couples (weighted $g^U = 0.006$, CI = −0.12–0.13) were significantly smaller than gender differences among noncouples (weighted $g^U = 0.19$, CI = 0.10–0.28), $Z(1) = 2.37$, $p < .025$. Because couples were also older than noncouples (mean age = 42.5 vs. 31.1, respectively), it was not possible to determine if the decline in gender differences among older adults reflects a maturational process or simply homogamous mating.

Sampling bias: adolescents. Nine samples were eliminated from the meta-analysis because they represented extreme effect sizes (outliers). Excluding these samples from the analysis may have biased the pattern of gender differences observed among adolescents. To examine this possibility, I repeated the meta-analysis with the outliers included in the computations. Results are presented in Table 4. Gender differences among adolescents were present, although they appeared to undergo a steady decline between the 9th and 12th grades. This decline contrasts sharply with the stability depicted in Figure 1. The difference in results is due to three effect sizes in the 12th-grade samples, which were considered outliers (weighted $g^U = -0.327, -0.013$, and 0.00). Their inclusion in the current analysis lowered the mean effect size for 12th graders and conveyed the impression of a steady decline in gender differences between the last years of high school and the first years of college. Inspection of the study characteristics associated with each outlier did not suggest what other variable (in addition to age) may have influenced the effect size.

Item sum versus Total Protocol Rating. Item-sum scoring was used more with adolescent samples than with adult samples. If this rating procedure is more sensitive to the presence of gender differences, then differences in scoring procedures might explain the smaller effect size found among college and adult samples.

To examine this issue, I located 23 samples for which effect sizes were reported for both item-sum and TPR data. Effect sizes were not significantly larger when computed from item-sum data (weighted $g^U = 0.31$) than when computed from TPR data (weighted $g^U = 0.25$), $Z = 0.87$, *ns.*

Discussion

As predicted, sex differences in development favored females, but this advantage was neither small nor uniform across the lifespan. Sex differences arose by late childhood; their presence in samples of young adolescents was unequivocal. Among 13-year-olds, male–female differences in personality development were moderately large, and they remained so for the next few years. Although young boys remained bound by egocentric concerns, young girls moved toward a period of social conformity; as boys entered a conformist period, girls approached a period of emerging self-awareness. Ironically, it is adolescent boys who are granted earlier dating privileges, independence, and freedom from adult supervision (Simmons & Blyth, 1987), although it is adolescent girls who display greater maturity of thought.

Sex differences in development declined among samples of older adolescents. The average effect size (g^U) among college students was small, less than one fourth of a standard deviation. This estimate was derived from a large and heterogeneous collection of student samples, reducing the likelihood that small gender differences characterize only a subset of college students (e.g., students at elite private universities).

Sex differences declined further among post-college-age adults and eventually disappeared. This, too, appears to be an unequivocal finding. Among couples, women displayed no greater maturity of thought than their male counterparts; among noncouples, women obtained slightly higher WUSCT scores than men, but the effect size was small, less than one fifth of a standard deviation.

The dramatic decline in sex differences between high-school and college samples could be due to a sampling bias. Presumably, male high-school students at low stages of ego development do not pursue college studies and would not be found among university samples. The absence of these males in college samples would automatically raise the mean ego level of men, thereby reducing the gap between male and female students. On the basis of this scenario, one would predict relatively small gender differences among samples of college-bound high-school students. Yet large sex differences were found among academically advanced 11th graders (Kitchener et al., 1984) and among 9th-, 10th-, and 11th-grade private school students identified as college bound (Hansell, 1981). Thus, sampling bias alone cannot account for the decline in gender differences observed among older adolescents.

The absence of sex differences among college upperclassmen and older adults most likely reflects maturational processes rather than biased sampling procedures. Holt's (1980) national probability sample of adults 22–25 years of age varied widely in educational level, social class background, and employment status, yet sex differences in ego development were not found in the sample (Browning, 1987), nor were differences observed among a representative sample of adults recruited from a small community in the midwestern United States (Hansell et al., 1985). Sex differences also appear to be absent among adults of low socioeconomic status (Suzman, 1974).

Decline in Sex Differences

When do sex differences begin to decline? Our results are equivocal on this issue. Some findings point to a steady decline between 9th and 12th grades (Table 4), but this evidence is

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Table 4
Mean Effect Sizes and Related Values, Broken Down by Age (Outlier Values Included)

Grade/age	Mean g^U (weighted)	Homogeneity statistic	Critical value[a]	No. of samples	N
5th–6th	−0.005	8.25*	7.81	4	330
7	0.26	20.20*	5.18	5	302
8	0.49	23.40*	14.07	8	848
9	0.72	11.77*	11.07	6	334
10	0.63	0.24	5.99	3	194
11	0.58	0.25	7.81	4	289
12	0.38	23.32*	14.07	8	525
Freshman	0.25	19.56	25.00	16	997
Sophomore–senior	0.13	9.29	15.51	9	386
22 < mean age < 35	0.21	5.62	16.92	10	1,174
38 < mean age < 74	0.03	13.85*	12.59	7	592

[a] Critical value of chi-square distribution of $n - 1$ degrees different and alpha = .05.
* $p < .05$.

unconvincing. The apparent decline is a result of including in the analysis three effect sizes that were judged to be outliers. Excluding these values from the analysis dramatically altered the pattern of sex differences. Rather than declining among high-school students, sex differences appeared relatively stable. The majority of evidence supports the latter interpretation.

The apparent stability of sex differences throughout adolescence is an unexpected finding. Boys and girls undergo dramatic personality changes between the 7th and 12th grades, yet girls seemed to maintain a constant lead over boys, suggesting that both sexes mature at the same rate until their late teens.

What factors account for the greater maturity observed in adolescent girls? Both sexes may confront the same pacers for development but react to them differently, or both sexes may encounter similar pacers, with the exposure for boys lagging 1–2 years behind the exposure for girls. Alternatively, both sexes may be exposed to the same pacers, at the same time, and with similar responses; the presence of sex differences may reflect a female growth spurt that occurs prior to adolescence.

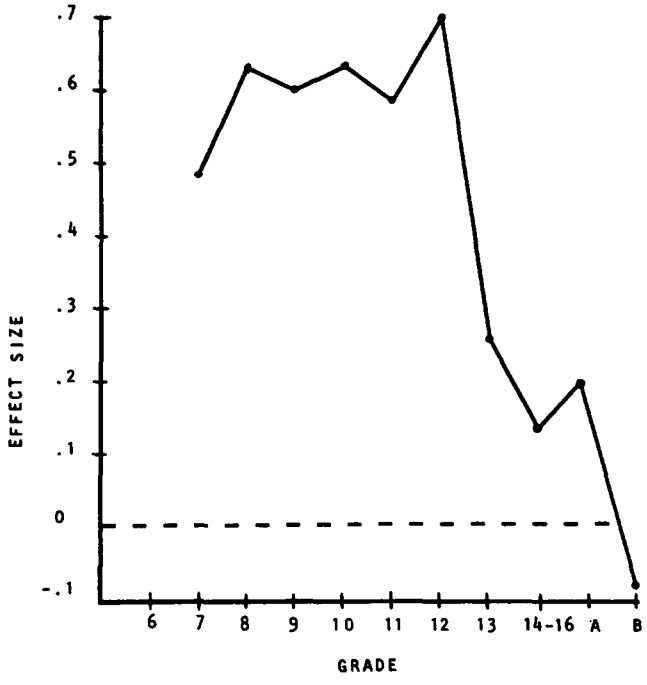

Figure 1. Magnitude of gender differences in ego development (weighted g^U, broken down by grade level). (Positive values indicate a female advantage; data were obtained from 67 samples [N = 5,071]; A = ages 23–35; B = ages 38–74.)

Roots of Sex Differences

Sex differences in personality development can be attributed to one of three factors: sex differences in cognitive abilities (in particular, verbal skills), sex differences in biological maturation, or sex differences in socialization experiences (Block, 1984). The first explanation is not supported by available evidence: Sex differences in ego development are at least twice the magnitude of sex differences in verbal ability. Hyde and Linn's (1988) review of 165 studies revealed no sex differences in vocabulary skills or reading comprehension, two skills that might influence responses to the WUSCT. The average effect size among all studies of verbal ability conducted prior to 1973 was only one fourth of a standard deviation, whereas the effect sizes among studies conducted after 1973 were smaller, approximately one tenth of a standard deviation. The majority of WUSCT studies were conducted after 1973, a period when male–female differences in verbal ability were virtually zero. Thus, sex differences in verbal capacities cannot account for the presence of sex differences in personality development.

The second explanation for sex differences in personality development (i.e., biological maturity) is more appealing because girls enter puberty approximately 2 years earlier than boys, beginning around age 11, that is, just about the time that gender differences in ego development begin to be observed. This explanation assumes that psychological maturity accompanies physiological maturity. However, psychological reactions to puberty depend on a number of factors, including one's gender

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and one's relative status among peers (e.g., whether one is an early, on-time, or late developer). Early developing girls, for example, may have the most difficulty adapting to puberty, whereas early developing boys may show the fewest problems (Brooks-Gunn & Petersen, 1983; Simmons & Blyth, 1987). With the exception of a single study of ego development and menarcheal age (Petersen & Crockett, 1985), the relationship of pubertal change to advances in ego development is unknown and requires additional study.

The third, and most likely, explanation for the rise of gender differences lies in the social experiences of boys and girls. Piaget (1932/1965) speculated that the loosely knit structure of girls' games accounts for their achievement of autonomous morality 2 years ahead of boys. Recent findings strengthen this conclusion and suggest how sex differences in social activity may also generate sex differences in personality development. Lever (1978) studied the spontaneous play of 183 fifth-grade children and found that boys played in large groups with little opportunity for discussion or conversation whereas girls played in small groups, often dyads, in which they mimicked basic human relationships. The latter activity may encourage the growth of impulse control and interpersonal awareness among girls, two characteristics associated with personality development.

Sex differences in personality are frequently attributed to socialization practices that encourage boys to value agentic traits (e.g., self-reliance, competitiveness, and achievement striving) and girls to value communal traits (e.g., gentleness, warmth, and sympathy). In contrast, Eagly (1987) proposed that sex differences in behavior arise from sex differences in the demands of male and female social roles. Neither explanation accounts for the sex differences examined in this review. The capacity to plan for the future and the capacity to adopt the perspective of another person represent developmental achievements that do not immediately appear or disappear by replacing one social expectation with another.

The impact of sex roles on personality development may not stem from sex roles themselves but from the meaning that children attach to these roles. Young boys, for example, may fail to distinguish between competitive and impulsive behavior, confusing parental encouragement of achievement behavior with parental sanctioning of impulsive behavior. The competitive orientation conferred on boys may simply reinforce an impulsivity that is present in all children.

Sex Differences in Ego Development and Moral Reasoning

Loevinger's WUSCT was initially validated on samples of females; Kohlberg's MJI was validated on samples of males. Despite these methodological biases, sex differences in both developmental systems appear to favor adolescent girls (Thoma, 1986; Walker, 1984), making it unlikely that girls' superior performance on the WUSCT is an artifact of scoring biases. The presence of sex differences on Loevinger's WUSCT, Rest's DIT, and, minimally, on Kohlberg's MJI is especially noteworthy. These assessment procedures vary widely in format, reducing the likelihood that shared method variance accounts for females' superior performance on all three measures.

Recent discussions suggest that *two* moral orientations exist: one orientation is more likely to be adopted by males and the other orientation is more likely to be adopted by females (Gilligan, 1982). The moral judgments of males are often guided by abstract moral principles and a concern with individual rights. The judgments of females are more contextual and oriented toward satisfying the competing needs of individuals in a given situation. From the standpoint of personality development, however, neither orientation can be viewed as more advanced than the other. Periods of egocentrism, social conformity, and self-evaluated judgments characterize the growth of moral reasoning within both Kohlberg's system and Gilligan's system.

Recent evidence suggests that personality development may reflect the capacity to use both moral perspectives. Rogers (1987) scored 394 WUSCT protocols obtained from males and females between the ages of 12 and 20. Protocols were scored for ego development, as well as for their degree of justice orientation (Kohlberg system) and care orientation (Gilligan system). As expected, protocols obtained from females contained the most care-oriented themes, and protocols obtained from males contained the most justice-oriented themes. However, as males and females advanced in ego level, they made increasing use of both moral orientations in their responses.

The progression from egocentrism to conformist reasoning and then on to self-evaluated judgments is descriptive of advances in moral reasoning, regardless of one's gender. A similar progression is descriptive of advances in personality development, supporting the belief that a single path toward maturity exists for both sexes.

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Appendix

Studies Used in the Meta-Analysis

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